NISQRC framework, quantum state evolution from time step $n-1$ to n can be written via the maps:

$$\hat{\rho}_n^{\mathrm{MR}} = \mathcal{U}(u_n)\left(\mathrm{Tr}_{\mathrm{R}}(\hat{\rho}_{n-1}^{\mathrm{MR}}) \otimes |0\rangle\langle 0|^{\otimes R}\right), \tag{8}$$

$$\hat{\rho}_n^{\mathrm{M}} = \mathrm{Tr}_{\mathrm{R}}(\mathcal{U}(u_n)(\hat{\rho}_{n-1}^{\mathrm{M}} \otimes |0\rangle\langle 0|^{\otimes R})) \equiv \mathcal{C}(u_n)\hat{\rho}_{n-1}^{\mathrm{M}}, \tag{9}$$

which describes the reset of the post-measurement readout subsystem after time step $n-1$, followed by input encoding via $\mathcal{U}(u_n)$ into the full quantum system state. With an eye towards the construction of an I/O map, it proves useful to introduce the expansion of the relevant single-step maps $\mathcal{U}(u)$ and $\mathcal{C}(u)$ in the basis of input monomials u^k: $\mathcal{U}(u)\hat{\rho}^{\mathrm{MR}} = \sum_{k=0}^{\infty} u^k \mathcal{R}_k \hat{\rho}^{\mathrm{MR}}$ and $\mathcal{C}(u)\hat{\rho}^{\mathrm{M}} = \sum_{k=0}^{\infty} u^k \mathcal{P}_k \hat{\rho}^{\mathrm{M}}$. Then, via iterative application of Eq. (9), $\hat{\rho}_n^{\mathrm{MR}}$ can be written as:

$$\hat{\rho}_n^{\mathrm{MR}} = \sum_{k_1,\cdots,k_n=0}^{\infty} u_1^{k_1} \cdots u_n^{k_n} \mathcal{R}_{k_n}\left(\mathcal{P}_{k_{n-1}} \cdots \mathcal{P}_{k_1}\hat{\rho}_0^{\mathrm{M}} \otimes |0\rangle\langle 0|^{\otimes R}\right). \tag{10}$$

The measured features $x_j(n)$ can then be obtained via $x_j(n) = \mathrm{Tr}(\hat{M}_j \hat{\rho}_n^{\mathrm{MR}})$.

In Supplementary Note 3, we show that these $x_j(n)$ obtained using the NISQRC framework can indeed be expressed as a Volterra series

$$x_j(n) = \sum_{k=0}^{\infty}\sum_{n_1=0}^{\infty}\cdots\sum_{n_k=n_{k-1}}^{\infty} h_k^{(j)}(n_1,\cdots,n_k)\prod_{\kappa=1}^{k} u_{n-n_\kappa} \tag{11}$$

in the infinite-shot limit. The existence of this manifestly time-invariant form is only possible due to the existence of an information steady-state, guaranteed for a quantum mechanical system under measurement.

Due to fading memory, the Volterra kernel $h_k^{(j)}(n_1,\cdots,n_k)$ characterizes the dependence of the systems' output at time n on inputs at most n_k steps in the past (recall $n_1 \le \cdots \le n_k$, see Eq. (11)). The evolution of $\hat{\rho}_n^{\mathrm{MR}}$ upto step $n-n_k$, namely for all $i < n-n_k$, is thus determined entirely by the null-input superoperator $\mathcal{P}_0$. Then the existence of a Volterra series simply requires the existence of an asymptotic steady state for the memory subsystem, $\lim_{n\to\infty}\mathcal{P}_0^n\hat{\rho}_0^{\mathrm{M}} = \hat{\rho}_{\mathrm{FP}}^{\mathrm{M}}$. As shown in the Supplementary Note 3, such a fixed point is usually ensured by the map $\mathcal{P}_0\hat{\rho}^{\mathrm{M}} = \mathcal{C}(0)\hat{\rho}^{\mathrm{M}} = \mathrm{Tr}_{\mathrm{R}}(\mathcal{U}(0)(\hat{\rho}^{\mathrm{M}} \otimes |0\rangle\langle 0|^{\otimes R}))$ being a CPTP map in generic quantum systems. This immediately indicates the fundamental importance of $\mathcal{P}_0$, the operator that corresponds to the single-step map of the memory subsystem under null input: it determines the ability of the NISQRC framework to evolve the quantum system to a unique statistical steady state, guaranteeing the asymptotic time-invariance property, and hence the existence of the Volterra series.

One byproduct of computing infinite-S features $\{x_j(n)\}$ is that it enables us to approximately simulate $\{\bar{X}_j(n)\}$ in a very deep N-layer circuit for finite S, without sampling individual quantum trajectories under N repeated projective measurements described in "Methods" subsection "Generating features via conditional evolution and measurement". In fact, given any n, once we evaluate a probability distribution $\{x_j(n) \ge 0\}$ satisfying $\sum_j x_j(n) = 1$, we can i.i.d. sample under this distribution vector for S shots and construct the frequency $\{\tilde{X}_j(n)\}$ as an approximation of $\{\bar{X}_j(n)\}$. The validity of this approximation is ensured by the additive nature of loss functions in the time dimension. More specifically, given Q input sequences $\{\mathbf{u}^{(q)} \in [-1,1]^N\}_{q\in[Q]}$, a general form of loss function is $\mathscr{L} = \frac{1}{QN}\sum_q\sum_n \mathcal{L}(\bar{\mathbf{X}}(n;\mathbf{u}^{(q)}))$. As shown in Appendix C5 of ref. 9, $\frac{1}{Q}\sum_q \mathcal{L}(\bar{\mathbf{X}}(n;\mathbf{u}^{(q)})) \approx \frac{1}{Q}\sum_q \mathcal{L}(\tilde{\mathbf{X}}(n;\mathbf{u}^{(q)}))$ in all orders of $\frac{1}{S}$-expansion for any $n \in [N]$, as long as Q is large enough. This is because the probability distribution of $\{\tilde{X}_j(n)\}$ is exactly the same as the distribution (marginal in time slice) of $\{\bar{X}_j(n)\}$. Therefore, $\frac{1}{QN}\sum_q\sum_n \mathcal{L}(\tilde{\mathbf{X}}(n;\mathbf{u}^{(q)}))$ is a good approximation of $\mathscr{L}$.

In Supplementary Note 2 and Supplementary Note 3, we show that without the reset operation, the fixed-point memory subsystem density matrix is the identity, $\hat{\rho}_{\mathrm{FP}}^{\mathrm{MR}} = \hat{I}^{\otimes L}/2^L$. While this steady state is independent of the initial state and therefore possesses a fading memory, it can be shown that the I/O map it enables is entirely independent of all past inputs as well, so that Volterra kernels $h_k^{(j)} = 0$ for any $k \le 1$. This yields a trivial reservoir, unable to provide any response to its inputs u. Such single-step maps $\mathcal{C}(u)$ are referred to as *unital maps* (maps that map identity to identity) and must be avoided for the NISQRC architecture to approximate any nontrivial functional. The inclusion of reset serves this purpose handily, although we have found certain improper encodings with reset to still result in unital maps $\mathcal{C}(u)$ (e.g., setting $n_T = 1$ in the circuit ansatz depicted in Fig. 4(a)).

A more rigorous sufficient condition for obtaining a nontrivial functional map, referred to as a *fixed-point non-preserving* map in the main text, is that $\mathcal{C}(u)$ does not share the same fixed points for all u. It is equivalently $\mathcal{P}_k\hat{\varrho}_{\mathrm{FP}}^{\mathrm{M}} \neq 0$ for some $k \ge 1$, due to the identity $\mathcal{C}(u)\hat{\varrho}_{\mathrm{FP}}^{\mathrm{M}} = \hat{\varrho}_{\mathrm{FP}}^{\mathrm{M}} + \sum_{k=1}^{\infty} u^k \mathcal{P}_k\hat{\varrho}_{\mathrm{FP}}^{\mathrm{M}}$. We will prove the importance of these criteria in Supplementary Note 3. The breaking of this criteria will lead to a memoryless reservoir for all earlier input steps: if $\mathcal{P}_k\hat{\rho}_{\mathrm{FP}}^{\mathrm{M}} = 0$ for all $k \ge 1$, then $h_k^{(j)}(n_1,n_2,\cdots,n_k) \neq 0$ only if $n_1 = n_2 = \cdots = n_k = 0$. A similar result for quantum reservoirs characterized by quantum channels can also be found from Theorem 2 in ref. 53.

Spectral theory of NISQRC: memory, measurement, and kernel structures

Recall that we can always define the spectral problem $\mathcal{P}_0\hat{\varrho}_\alpha^{\mathrm{M}} = \lambda_\alpha\hat{\varrho}_\alpha^{\mathrm{M}}$ where $\hat{\varrho}_\alpha^{\mathrm{M}}$ are eigenvectors that exist in the $(2^M)^2 = 4^M$-dimensional space of memory subsystem states and whose eigenvalues satisfy $1 = \lambda_1 \ge |\lambda_2| \ge \cdots \ge |\lambda_{4^M}| \ge 0$. The importance of the spectrum of $\mathcal{P}_0$ is obvious from the definition of $\hat{\rho}_{\mathrm{FP}}^{\mathrm{M}}$ already. As $\hat{\rho}_{\mathrm{FP}}^{\mathrm{M}}$ is the fixed point of the map defined by $\mathcal{P}_0$, it must equal the eigenvector $\hat{\varrho}_1^{\mathrm{M}}$ since $\lambda_1 = 1$. Then writing the initial density matrix in terms of these eigenvectors, $\hat{\rho}_0^{\mathrm{M}} = \sum_\alpha d_{0\alpha}\hat{\varrho}_\alpha^{\mathrm{M}}$, the fixed point becomes $\hat{\rho}_{\mathrm{FP}}^{\mathrm{M}} = \lim_{n\to\infty}\left(\hat{\varrho}_1^{\mathrm{M}} + \sum_{\alpha\ge 2} d_\alpha^0 \lambda_\alpha^n \hat{\varrho}_\alpha^{\mathrm{M}}\right)$. This not only reproduces the result $\lim_{n\to\infty}\mathcal{P}_0^n\hat{\rho}_0^{\mathrm{M}} = \hat{\rho}_{\mathrm{FP}}^{\mathrm{M}}$ but also shows that the approach to the fixed point $\hat{\rho}_{\mathrm{FP}}^{\mathrm{M}} = \hat{\varrho}_1^{\mathrm{M}}$ must be determined by the magnitude of λ_2; the smaller the magnitude, the faster terms for $\alpha \ge 2$ decay and hence, the shorter the memory time.

To see more directly how the spectrum of $\mathcal{P}_0$ influences the memory of inputs, it is sufficient to analyze the Volterra kernels in Eq. (1). Focusing on single-time contributions from u_{n-p} to $x_j(n)$ at all orders of nonlinearity (multi-time contributions are exponentially suppressed, see Supplementary Note 4), these may be expressed as

$$\sum_{k=1}^{\infty} h_k^{(j)}(p^{\otimes k})\, u_{n-p}^k = \sum_{\alpha=2}^{4^M} \nu_\alpha^{(j)} \lambda_\alpha^{p-1} F_\alpha(u_{n-p}), \tag{12}$$

which can be viewed as a spectral representation of Volterra kernel contributions to the jth measured feature obtained via POVM $\hat{M}_j$. Here, $F_\alpha(u) = \sum_{k=1}^{\infty} c_{\alpha 1}^{(k)} u^k$ define $4^M - 1$ *internal features*, so-called as they depend only on input encoding operators via $\mathcal{P}_k\hat{\varrho}_{\alpha'}^{\mathrm{M}} = \sum_{\alpha=2}^{4^M} c_{\alpha\alpha'}^{(k)}\hat{\varrho}_\alpha^{\mathrm{M}}$, and are in particular independent of the measurement scheme. Nontrivial $F_\alpha(u)$ and $c_{\alpha 1}^{(k)}$ can be guaranteed if $\mathcal{P}_k\hat{\varrho}_{\mathrm{FP}}^{\mathrm{M}} \neq 0$ for some $k \ge 1$. The dependence of observables on the measurement basis is via coefficients $\nu_\alpha^{(j)} = \mathrm{Tr}(\hat{M}_j\mathcal{R}_0(\hat{\varrho}_\alpha^{\mathrm{M}} \otimes |0\rangle\langle 0|^{\otimes R}))$. Crucially, the weighting of $F_\alpha(u_{n-p})$ for p steps in the past is determined by eigenvalues λ_α^{p-1} of $\mathcal{P}_0$. For each $\alpha \ge 2$, it vanishes when we take a long time limit $p \to \infty$. This property is usually referred to as *fading memory*. It also clearly defines a set of distinct but calculable, memory fading rates $\{|\lambda_\alpha|\}_{\alpha\ge 2}$.

Importantly, the ability to construct Volterra kernels and internal features enables us to approximately treat the infinite-dimensional function $x_j(n) = \mathcal{F}_j(u_{\le n})$ as a function with support only over a space